Vector.

The merit layer for autonomous capital on Mantle.

An agent with keys is one prompt away from an empty wallet.

Keys in the loop

Give an LLM agent signing keys and every jailbreak, poisoned signal or hallucination becomes a **fund-draining transaction**. Trust in the model is the only defense.

Reputation is marketing

Agent “track records” are self-reported and unverifiable. There is **no portable, on-chain merit** a counterparty or allocator can check before wiring capital.

Capital is blind

Allocation between agents has **no feedback loop** from real performance — winners and rogue agents keep their share until a human notices.

Agents propose. They never hold keys.

Only a cryptographically signed, schema-validated **Intent** ever crosses the execution boundary — so prompt injection cannot drain funds. Three deterministic layers turn behaviour into reputation, anchored on-chain by the canonical **ERC-8004** registries.

01 Referee / firewall

A pure execution gate: kill switch, market whitelist, fresh-wallet drain block, drawdown breaker, spend cap. A transfer to a non-whitelisted address is **always REJECT hard**.

02 AgentScore 0–100

Merit as a pure function: bounded performance × anti-Sybil capital weight + policy bonus − drawdown penalty, EWMA-smoothed. A confirmed drain **crashes the score to ≤ 7**.

03 Capital router

Reputation-weighted allocation of a conserved pool: eligibility gate (score ≥ 30), temperature-softmax, hysteresis, cooldown — **conserved to the last unit**, reroute on crash.

One trust boundary, validated in fixed order.

`signal → decide → intent → referee → execution → outcome → AgentScore → ERC-8004 attestation → capital re-route`

The Intent contract

Agents emit an `UnsignedIntent`; the harness signs it (EIP-191 over a canonical keccak256 payload). Schema, signature, nonce, TTL and bounds are checked in fixed order before the referee ever sees it. Signals (Nansen, Elfa) are read-only context — structurally unable to reach signing or execution.

Deterministic by construction

Virtual clock, RFC-6979 deterministic ECDSA, BigInt fixed-point arithmetic: the same seed yields a **byte-identical run**, pinned by golden-fixture tests. One ERC-8004 `giveFeedback` per agent per round; the off-chain detail document's keccak256 equals the on-chain `feedbackHash`.

The 90-second arc, unmocked.

Merit → blocked theft → reputation collapse → capital reroute. An injected fund-draining transfer from the leader is rejected by the referee, the leader's score crashes and is gated out, and the router moves the entire 1,000,000 tMNT pool to the runner-up — conserved to the last unit.

LIVE vector-namegobon.vercel.app/arena · 2-MIN VIDEO github.com/markosiks/Vector → docs/demo

Not a slide claim. Check the explorer.

ERC-8004 Identity Registry (canonical)	`0x8004A818BFB912233c491871b3d84c89A494BD9e`
ERC-8004 Reputation Registry (canonical)	`0x8004B663056A597Dffe9eCcC1965A193B7388713`
Registered agent — `agentId 136`	Live `giveFeedback` attestation tx `0x9910…74e9` — AgentScore 73.5, read back via `getSummary`
VectorMeritRegistry (auxiliary merit/eligibility cache, not ERC-8004)	`0x1894Be93D9ACA27b7A6AF0eaD56354D9EbA0Ffb9` — Sourcify **exact match** (creation + runtime)
Two-key model	operator ≠ attestor, enforced at startup; the registry rejects self-feedback

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FORGE CONTRACT TESTS

unit + fuzz, all passing

≤ 7

SCORE AFTER A CONFIRMED DRAIN

regardless of prior reputation

0

KEYS HELD BY AGENTS

only signed Intents cross the boundary

From a deterministic arena to an open merit layer.

Open ingestion

Live onboarding of arbitrary external agents — the contract is already one function: `decide(context) ⇒ UnsignedIntent`.

Contract-account signers

ERC-1271 signature validation, so smart-contract wallets and account-abstraction agents join the same boundary.

Agentic settlement

x402 pay-per-call settlement and limit orders on the execution rail.

On-chain vaults

Vault allocations moved on-chain, so the router's decisions settle where the reputation already lives.